ICT303 – Advanced Machine Learning and Artificial Intelligence

Topic 2: Preliminaries

Hamid Laga
H.Laga@murdoch.edu.au
Office: 245.1.020

How to Get in Touch with the Teaching Team

- Internal and External Students
  - Email: H.Laga@murdoch.edu.au.
- Important
  - In any communication, please make sure that you
    - Start the subject of your email with ICT303
    - Include your student ID, name, and the lab slot in which you are enrolled.
  - We will do all our best to answer your queries within 24 hrs.

In this Lecture

- Introduction to Linear Algebra
  - Points, vectors, matrices
  - Dot product, matrix operations
- Probability and Statistics
  - Random variables,
  - Conditional probabilities
  - Probability distributions
  - Bayes rule
- Gradients
  - Definition, chain rile, automatic differentiation
- Summary

- Learning objectives
  - Overview of the main mathematical concepts needed for the unit
  - Familiarize yourself with Python, NumPy and PyTorch
  - Gradient and automatic differentiation
- Additional readings
  - Chapter 2 (Preliminaries) of the textbook, available at: https://d2l.ai/

Linear Algebra - Scalars

- A scalar is a just a number
- Simple operations

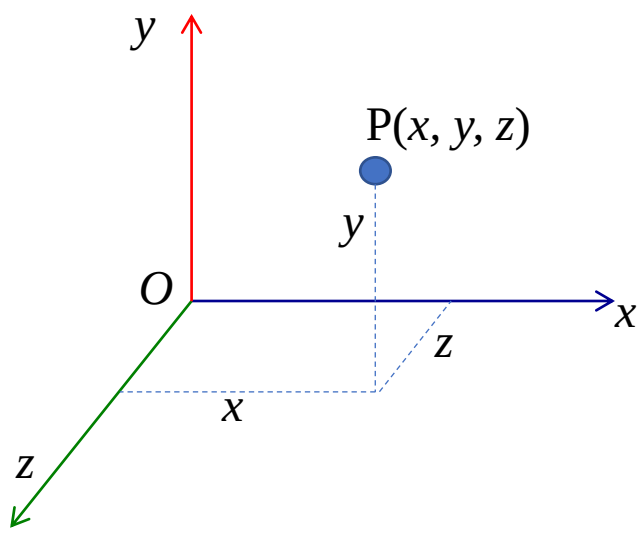
$$c = a + b$$
$$c = a \cdot b$$
$$c = \sin a$$

- Length (or norm)
  - The length of scalar is its absolute value

$$|a| = \begin{cases} a & \text{if} a > 0 \\ -a & \text{otherwise} \end{cases}$$
$$|a + b| \leq |a| + |b|$$
$$|a \cdot b| = |a| \cdot |b|$$

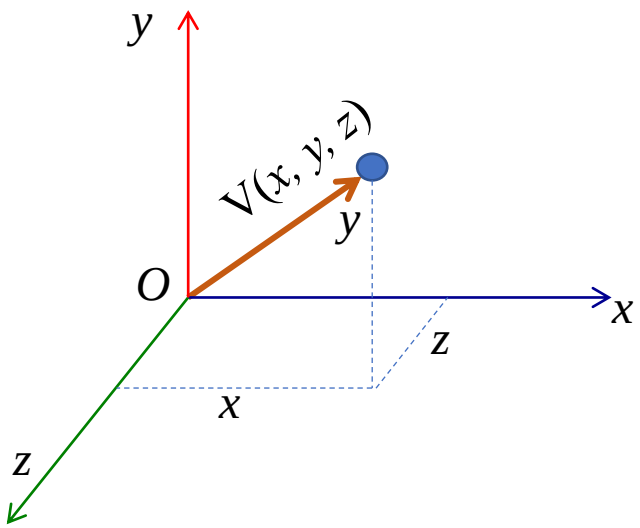
Linear Algebra - Points and Vectors

Points



- A point in the 3D space is represented with coordinates measured with respect to an origin O and the X, Y, Z axes

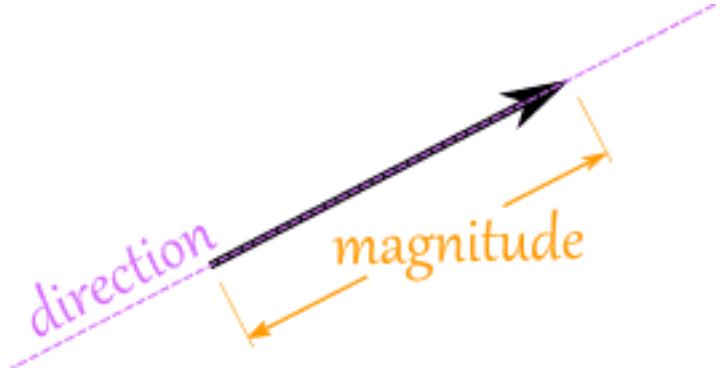
Vectors



- A vector represents a direction in space, with respect to a set of X, Y, Z axes.
- It has a magnitude called also length or norm.
- A unit vector is a vector of length 1

A vector has a direction and a magnitude

- It does not have an origin



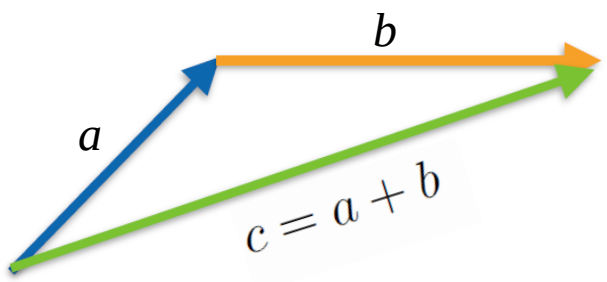
The magnitude, length or norm of a vector

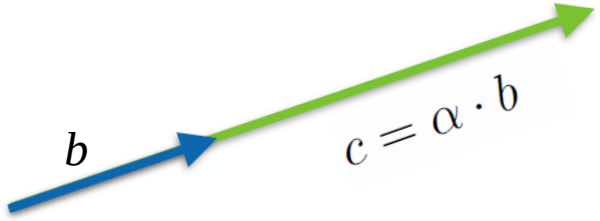
$$\|V\| = \sqrt{x^2 + y^2 + z^2}.$$

Linear Algebra - Vectors

- Simple operations

$$c = a + b \quad \text{where } c_i = a_i + b_i$$
$$c = \alpha \cdot b \quad \text{where } c_i = \alpha b_i$$
$$c = \sin a \quad \text{where } c_i = \sin a_i$$





- Length (norm)

$$\|a\|_2 = \left[\sum_{i=1}^{m} a_i^2\right]^{\frac{1}{2}}$$
$$\|a\| \geq 0 \text{ for all } a$$
$$\|a + b\| \leq \|a\| + \|b\|$$
$$\|a \cdot b\| = |a| \cdot \|b\|$$

Linear Algebra - Operations Vectors

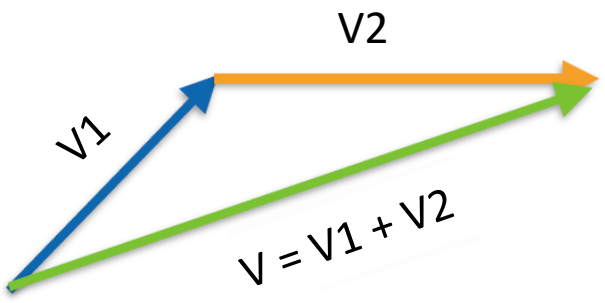
Let $V_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$ and $V_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$

be two vectors or points in the 3D space and α a scalar value, then

Linear Algebra - Operations Vectors

- Adding two vectors (addition)

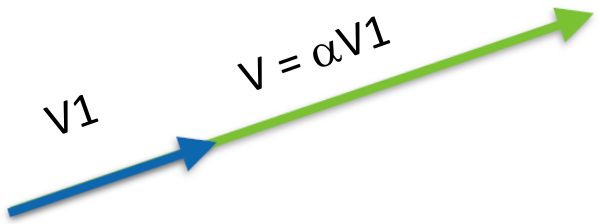
$$V = V_1 + V_2 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \end{bmatrix}$$



Linear Algebra - Operations Vectors

- Product of a scalar (number) with a vector

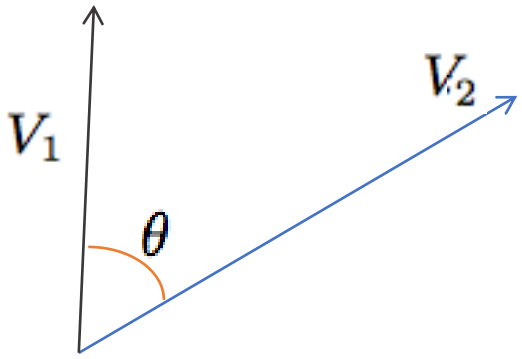
$$V = \alpha V_1 = V_1 \alpha = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \alpha x_3 \end{bmatrix}$$



Linear Algebra - Operations Vectors

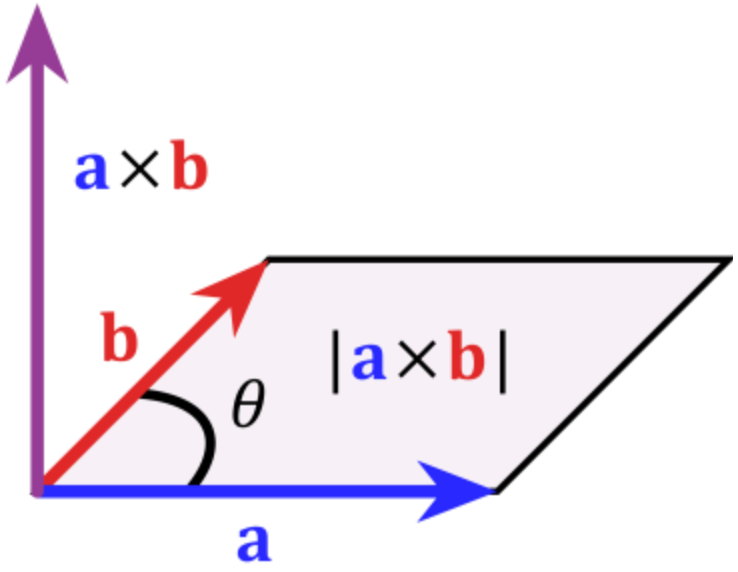
- Dot (inner) product of two vectors

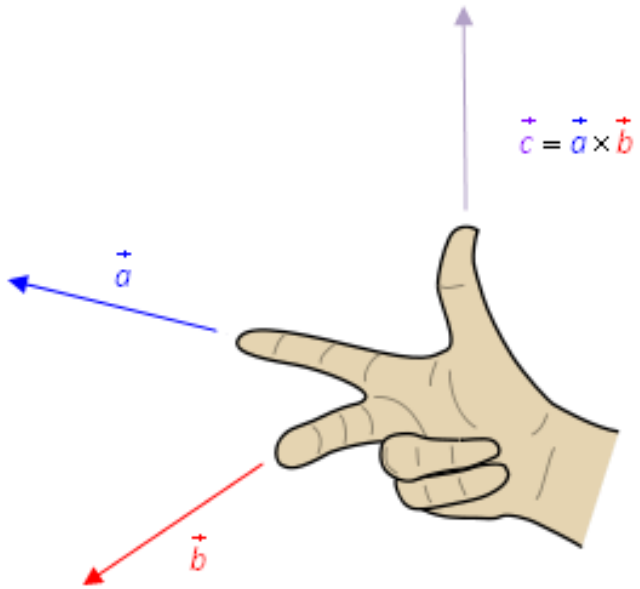
$$
\begin{aligned}
V_1 \cdot V_2 &= \langle V_1, V_2 \rangle = \|V_1\| \|V_2\| \cos\theta \\
&= x_1 x_2 + y_1 y_2 + z_1 z_2.
\end{aligned}
$$



Linear Algebra - Operations Vectors

- Cross (outer) product of two vectors





$$\|V\| = \|V_1\|\|V_2\| \sin\theta.$$

$$V = \begin{bmatrix} y_1 z_2 - z_1 y_2 \\ z_1 x_2 - x_1 z_2 \\ x_1 y_2 - y_1 x_2 \end{bmatrix}$$

Linear Algebra - Matrices

- A matrix is a two-dimensional array
  - It has a certain number of rows and columns

$$M = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mk} \end{bmatrix}$$

- Special cases
  - A vector is a matrix which has one column.
  - A scalar is also a matrix which has one column and one row

Linear Algebra - Matrices

- A matrix is a two-dimensional array
  - It has a certain number of rows and columns

$$M = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mk} \end{bmatrix}$$

- Special cases
  - A vector is a matrix which has one column.
  - A scalar is also a matrix which has one column and one row

- Matrices also have a norm
  - Frobenus norm of a matrix

$$\|A\|_{\text{Frob}} = \left[\sum_{ij} A_{ij}^2 \right]^{\frac{1}{2}}$$

Linear Algebra - Matrices

- Square matrix
  - A matrix that has the same number of columns and rows.
- Identity matrix
  - A square matrix whose diagonal elements are all 1, and its off-diagonal elements are 0

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2 x 2

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3 x 3

Linear Algebra - Matrices

- Addition of matrices
  - Given two matrices A, B of the same size

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix}, B = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{bmatrix},$$

then

$$C = A+B = B+A = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \dots & a_{1m}+b_{1m} \\ a_{21}+b_{21} & a_{22}+b_{22} & \dots & a_{2m}+b_{2m} \\ \dots & & & \\ a_{n1}+b_{n1} & a_{n2}+b_{n2} & \dots & a_{nm}+b_{nm} \end{bmatrix}.$$

Linear Algebra - Matrices

- Product of a scalar (number) with a matrix A

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix}, \qquad C = \alpha A = \begin{bmatrix} \alpha a_{11} & \alpha a_{12} & \dots & \alpha a_{1m} \\ \alpha a_{21} & \alpha a_{22} & \dots & \alpha a_{2m} \\ \dots & & & \\ \alpha a_{n1} & \alpha a_{n2} & \dots & \alpha a_{nm} \end{bmatrix}$$

Linear Algebra - Matrices

- Matrix transpose (or the transpose of a matrix
  - If A is a matrix of size m x n, then its transpose is a matrix C = A^t of size n x m, and defined as follows

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \Rightarrow C = A^t = \begin{bmatrix} a_{11} & a_{21} & \ldots & a_{m1} \\ a_{12} & a_{22} & \ldots & a_{m2} \\ \ldots & & & \\ a_{1n} & a_{2n} & \ldots & a_{mn} \end{bmatrix}.$$

Linear Algebra – Special Matrices

- **Symmetric, antisymmetric** $A_{ij} = A_{ji}$ and $A_{ij} = -A_{ji}$

- **Positive definite**

$$\|x\|^2 = x^\top x \geq 0 \text{ generalizes to } x^\top A x \geq 0$$

(all positive eigenvalues)

Linear Algebra - Matrices

- Symmetric matrix
  - A square matrix A is said symmetric if it is equal to its transpose

$$A^t = A.$$

  - In other words, $A_{ij} = A_{ji}$

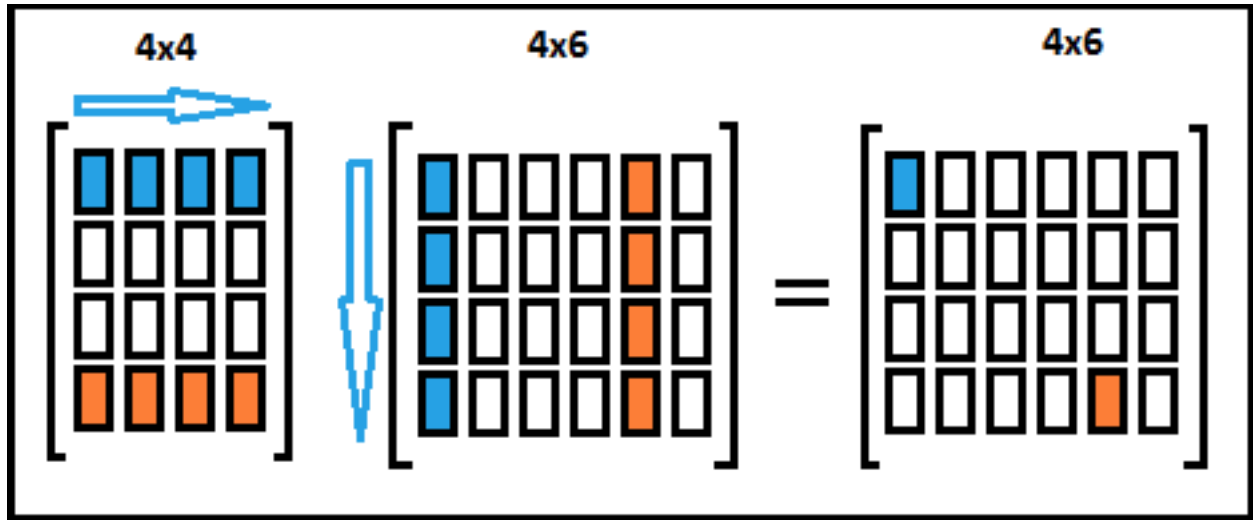
- Antisymmetric matrix

$$A_{ij} = -A_{ji}$$

Linear Algebra - Matrices

- Product of matrices
  - Let A be a matrix of size n x m
  - Let B be a matrix of size m x s
  - Then C = A x B is another matrix of size n x s.
  - The elements of C are defined as follows

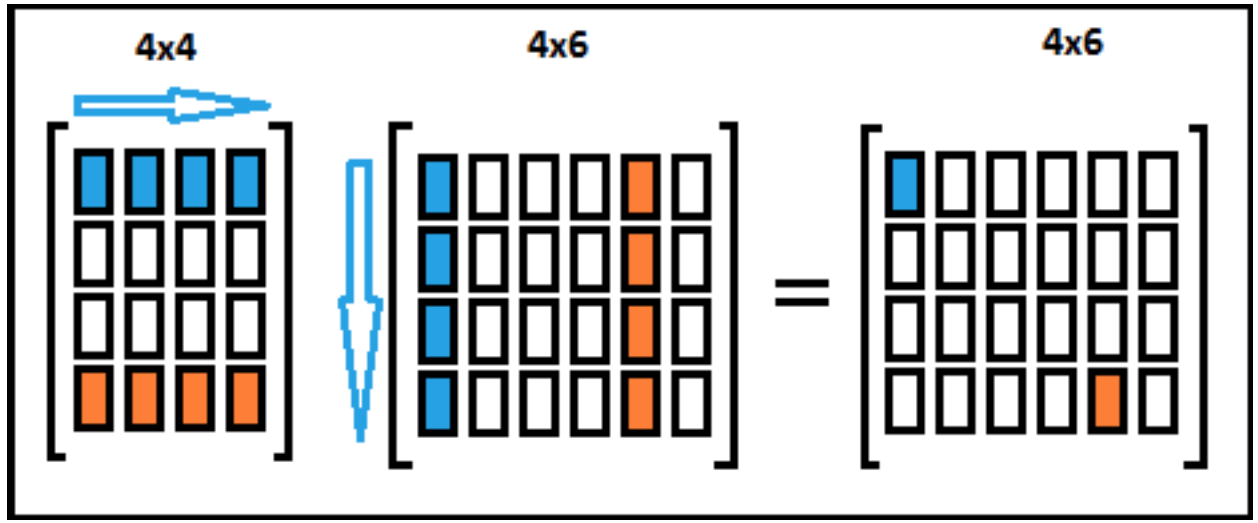
$$c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}$$



Linear Algebra - Matrices

- Product of matrices
  - Let A be a matrix of size n x m
  - Let B be a matrix of size m x s
  - Then C = A x B is another matrix of size n x s.
  - The elements of C are defined as follows

$$c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}$$



- The matrix product is not permutative

$$A \times B \neq B \times A$$

Linear Algebra - Matrices

- Eigenvectors and eigenvalues
  - Eigenvectors are vectors that are not changed by the matrix

$$Ax = \lambda x$$

  - For symmetric matrices, we can always find this.
- Discuss the geometric meaning of eigenvectors and eigenvalues
- Discuss the statistics meaning of eigenvectors and eigenvalues

Linear Algebra - Matrices

- Orthogonal matrices
  - All rows of the matrix are orthogonal to each other (i.e., their dot product is 0)
  - All rows of the matrix have unit length (i.e., their norm/length is equal 0)
  - In other words, a matrix U is orthogonal if and only if

$$UU^\top = \mathbf{1}$$

Note

- GPUs love vectors and matrices

Probability and Statistics

- Basic Probability
  - Random variables
  - Conditional probabilities
  - Bayes rule
  - Naïve Bayes
- Distributions
  - Normal (Gaussian) distribution
  - Central limit theorem

Basic Probability – Random Variables

- A Random variable, denoted X, is a numerical description of the outcome of a (statistical) experiment (from Britannica)
  - Daily temperature
  - Daily share price
  - Student mark in ICT303
- A random variable can be
  - Discrete, e.g., the species of the animal observed in an image is a random variable that takes values in {dog, cat, horse, ant, spider, …}
  - Continuous, e.g., daily temperatures

Basic Probability – Probability & Probability Distribution

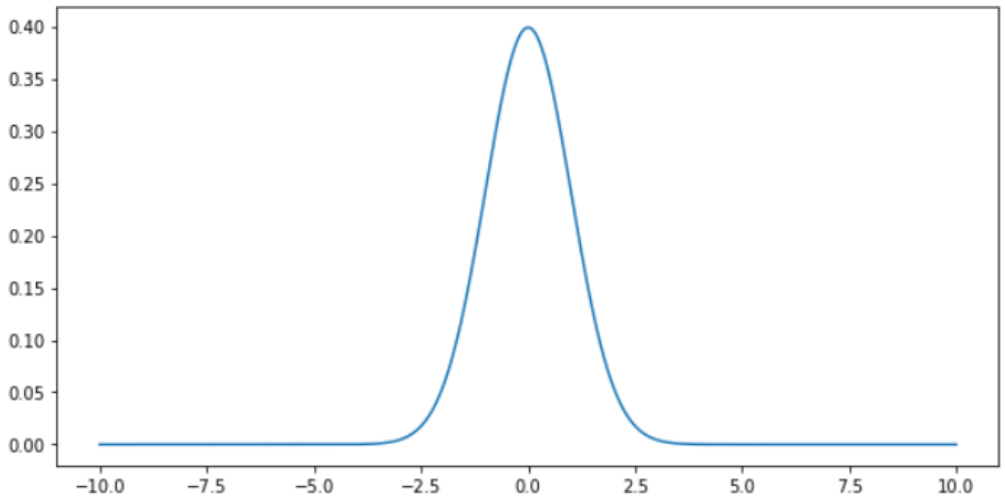
- Probability
  - Let X be a random variable describing the mark of a student in ICT303
  - Pr(X = 50) is the probability of X to be equal to 50
    - In other words, it describes the chance of obtaining 50 in the unit
  - Pr(X>=50) is the probability of passing the unit
  - Pr(X<50) is the probability of failing the unit

- Probability distribution for a random variable
  - Describes how the probabilities are distributed over the values of the random variable
  - Example – ICt303,
    - the mark values are between 0 and 100.
    - What is the chance of having 0, 1, 2, …., 100 in this unit?

Basic Probability – Probability & Probability Distribution

- Example: Normal (Gaussian) Distribution

  - Density $$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



Mean μ Variance σ^2 Standard deviation σ

Basic Probability

- Central Limit Theorem
  - The random variable below converges to a Gaussian

$$z_n := \left[\sum_{i=1}^{n} \sigma_i^2\right]^{-\frac{1}{2}} \left[\sum_{i=1}^{n} x_i - \mu_i\right]$$

  - Practical application
    - Sum of independent random variables will converge to a normal distribution with joint variance

Basic Probability

- Sometimes we have many random variables,
  - e.g., weather can be described with two random variables
    - X: temperature
    - Y: rain fall
- Joint Probabilities
  - What is the probability of having X = 35 and Y = 30? → Pr(X = 35 and Y = 30) which is equivalent to Pr(X = 35, Y = 30)
- Conditional Probabilities
  - Let Z be the location (e.g., Perth, Darwin, Melbourne, Sydney, etc.)
  - What is the probability of having
    - X = 35 and Y = 30 given that the location is Darwin?
    - X = 35 and Y = 30 given that the location is Sydney?
    - We can write it as Pr(X=35, Y=30 | Z = “Sydney”)

Basic Probability

- (In)dependence
  - Two random variables can be dependent on each other
    - Rain and temperature
    - Petrol prices and care sales
  - Two random variables can be independent
    - Your mark in ICT303 and temperature in NY

Basic Probability – Joint Probability and Bayes Rule

- Joint probability of two random variables

$$\Pr(X, Y) = \Pr(X|Y)\Pr(Y) = \Pr(Y|X)\Pr(X)$$

- Bayes rule

$$\Pr(X|Y) = \frac{\Pr(Y|X) \cdot \Pr(X)}{\Pr(Y)}$$

Basic Probability – Joint Probability and Bayes Rule

- Application - Spam Filter
  - Key assumption
    - Words occur independently of each other given the label of the document (here, label means spam or not spam)

$$p(w_1, \ldots, w_n | \text{spam}) = \prod_{i=1}^{n} p(w_i | \text{spam})$$

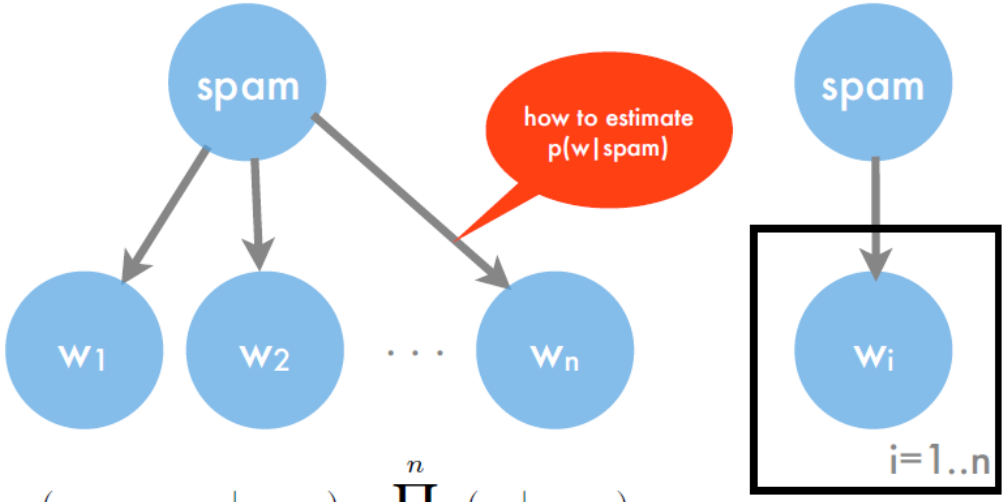
  - Spam classification via Bayes rule

$$p(\text{spam} | w_1, \ldots, w_n) \propto \underbrace{p(\text{spam})}_{\text{Spam probability}} \prod_{i=1}^{n} \underbrace{p(w_i | \text{spam})}_{\text{Distribution of words in spam emails}}$$

Basic Probability – Joint Probability and Bayes Rule

- Application - Spam Filter
  - Key assumption
    - Words occur independently of each other given the label of the document (here, label means spam or not spam)

$$p(w_1, \ldots, w_n | \text{spam}) = \prod_{i=1}^{n} p(w_i | \text{spam})$$



$$p(w_1, \ldots, w_n | \text{spam}) = \prod_{i=1}^{n} p(w_i | \text{spam})$$

Graphical representation

Basic Probability – Joint Probability and Bayes Rule

- Application - Spam Filter
  - Key assumption
    - Words occur independently of each other given the label of the document (here, label means spam or not spam)

$$p(w_1, \ldots, w_n | \text{spam}) = \prod_{i=1}^{n} p(w_i | \text{spam})$$

  - Spam classification via Bayes rule

$$p(\text{spam} | w_1, \ldots, w_n) \propto \underbrace{p(\text{spam})}_{\text{Spam probability}} \prod_{i=1}^{n} \underbrace{p(w_i | \text{spam})}_{\text{Distribution of words in spam emails}}$$

Basic Probability

- Application - Spam Filter
  - Data
    - Emails (headers, body, metadata)
    - Labels (spam/non-spam) – assume that users actually label all mails
    - …
  - Need to estimate
    - Probability of spam: P(spam), and
    - Distribution of words in spams: P(wi | spam)

- Simple algorithm

- For each document (x,y) do
  - Aggregate label counts given y
  - For each feature x_i in x do
    - Aggregate statistic for (x_i, y) for each y
- For y estimate distribution p(y)
- For each (x_i,y) pair do
  Estimate distribution p(x_i|y), e.g. Parzen Windows, Exponential family (Gauss, Laplace, Poisson, ...), Mixture

- Given new instance compute $p(y|x) \propto p(y) \prod_j p(x_j|y)$

Summary

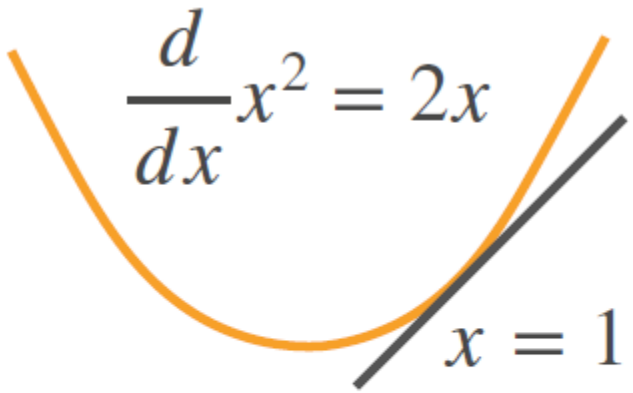
- Basic Probability
  - Random variables, conditional probabilities, Bayes rule, distribution, central limit theorem
- Gaussian distribution is very important

Gradients

- Gradients
  - Definition and what is it for
  - Chain rule
  - Automatic differentiation

Derivatives and Gradients

- Derivative of a function with one variable
  - It is the rate of change in a function with respect to changes in its arguments (parameters)
  - Geometrically, it is the slope of the tangent line



The slope of the tangent line is 2

Derivatives and Gradients

- Partial derivatives and gradients

- Let $f(x_1, x_2, \ldots, x_n)$ be a function of n variables
  - The partial derivative of f with respect to a variable x_i is the derivative of f when all the variables, except x_i, are considered as constants
  - Example:
  - It is denoted by $\frac{\partial f}{\partial x_i}$ or $\partial_{x_i} f$

- The gradient of f with respect to all the variables $x = (x_1, \ldots, x_n)$ is

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = [\partial_{x_1} f(\mathbf{x}), \partial_{x_2} f(\mathbf{x}), \ldots \partial_{x_n} f(\mathbf{x})]^\top.$$

Derivatives and Gradients – Where do I need them?

- Gradients are used to optimize functions
  - Given a function of a certain no. of parameters, we would like to find the values of the parameters for which the function is maximum (or minimum)
  - Many problems can be formulated that way
    - Hedging your share portfolio
    - Finding the 3D body shape that corresponds to the one seen in image
    - Etc.
    - Training (Deep) Neural Networks is the process of finding the network parameters that give minimum error (mistakes) on the training data

- Calculating derivatives / gradients is crucial to any learning algorithm

Gradients – Automatic Differentiation (AD)

- AD evaluates gradients of a function specified by a program at given values
  - It is also called Autograd
  - It is implemented in many deep learning frameworks such as PyTorch, MxNet, TensorFlow

- How is it done?
  - As we pass data through each successive function, the framework builds a computation graph that tracks how each value depends on others
  - Automatic differentiation package then works backwards through the graph applying the chain rule

Gradients – Automatic Differentiation (AD)

- Example 1 – Simple function
  - Let x be a column vector of n variables
  - Let the function $y = 2\mathbf{x}^\top \mathbf{x}$
  - We want to calculate the gradient of the function with respect to x.

- Please refer to the notebook associated with this lecture topic

https://d2l.ai/chapter_preliminaries/autograd.html

This week’s lab

- Continue with last week’s exercises
- Practice, with Python, some of the concepts we introduced in this lecture

Next Week

- Linear Neural Networks

Questions

Basic Probability – Joint Probability and Bayes Rule

- Bayes Rule – e.g., AIDS test
  - Data
    - Approximately 0.1% are infected
    - AIDS test detects all infections
    - AIDS test reports positive from 1% healthy people
  - What is the probability of having AIDS if test is positive

$$
\begin{aligned}
\Pr(a=1|t) &= \frac{\Pr(t|a=1)\cdot\Pr(a=1)}{\Pr(t)} \\
&= \frac{\Pr(t|a=1)\cdot\Pr(a=1)}{\Pr(t|a=1)\cdot\Pr(a=1)+\Pr(t|a=0)\cdot\Pr(a=0)} \\
&= \frac{1\cdot 0.001}{1\cdot 0.001+0.01\cdot 0.999} = 0.091
\end{aligned}
$$

Gradients – Chain Rule

- Chain rule for scalars:

$$y = f(u),\ u = g(x) \qquad \frac{\partial y}{\partial x} = \frac{\partial y}{\partial u}\frac{\partial u}{\partial x}$$

- Generalize to vectors straightforwardly

$$\frac{\partial y}{\partial \mathbf{x}} = \frac{\partial y}{\partial u}\frac{\partial u}{\partial \mathbf{x}} \qquad \frac{\partial y}{\partial \mathbf{x}} = \frac{\partial y}{\partial \mathbf{u}}\frac{\partial \mathbf{u}}{\partial \mathbf{x}} \qquad \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \frac{\partial \mathbf{y}}{\partial \mathbf{u}}\frac{\partial \mathbf{u}}{\partial \mathbf{x}}$$

$$(1,n) \quad (1,)\ (1,n) \qquad (1,n) \quad (1,k)\ \ (k,n) \qquad (m,n)\ \ (m,k)\ (k,n)$$